



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO CIBI PANDIYAN AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

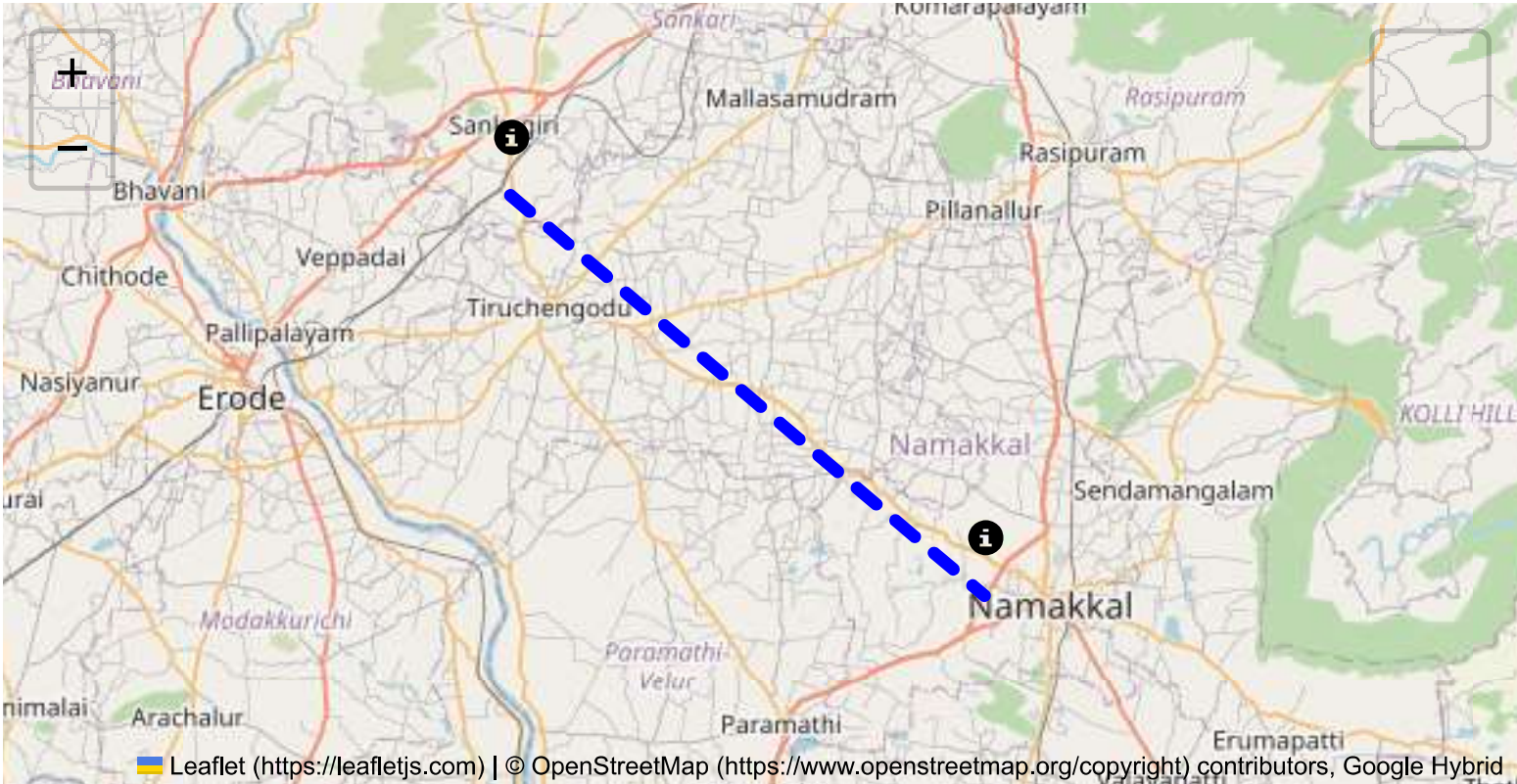
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

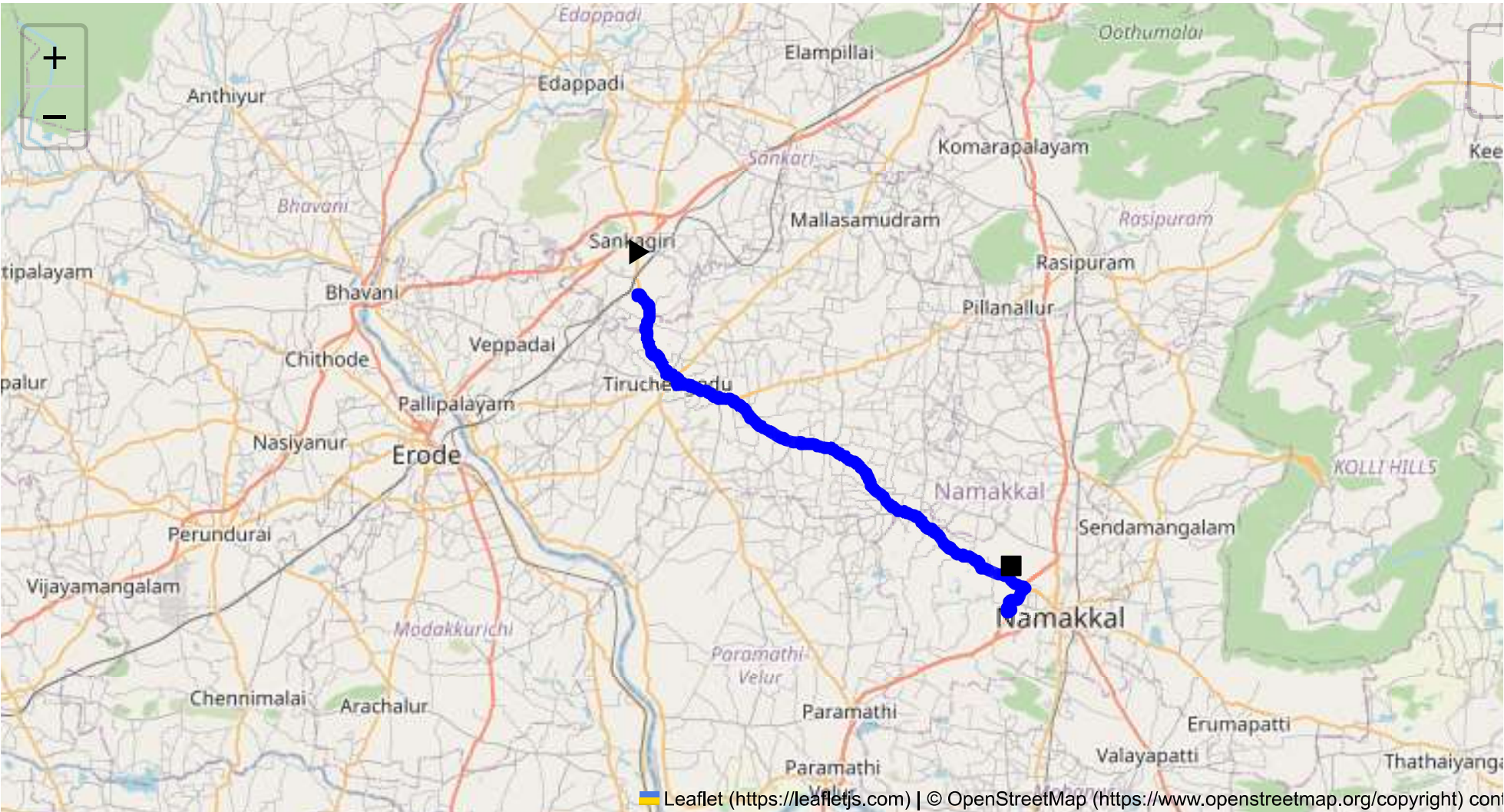
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 43.37 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.224816, 78.132652)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Namakkal via Route 637301 to 637003 covers approximately 43.37 kilometers. Starting in Sangagiri, the route likely involves traveling on National Highways and local roads that connect these two locations. Details of exact roads and intersections would be necessary for a complete map analysis, including potential toll plazas and notable junctions.

Typical Weather Conditions and Potential Hazards

Tamil Nadu generally experiences a tropical climate with hot, dry summers and wet, humid monsoons. During the monsoon season (June to December), heavy rainfall can cause flooding and reduce visibility. Summers can lead to heatwaves and road surface wear. Weather-related hazards include slippery roads and potential flooding, especially in low-lying areas.

Traffic Patterns and Congestion

The traffic patterns along this route are typically influenced by local commuting hours and commercial transport schedules. Expect congestion during peak hours, typically in the mornings (8:00–10:00 AM) and evenings (5:00–7:00 PM). Major towns, market areas, and intersections are prone to traffic jams, with vehicle breakdowns potentially exacerbating congestion.

Road Quality and Infrastructure

Generally, the national highways in Tamil Nadu are well-maintained, but local roads and connecting routes might have uneven surfaces, potholes, or narrower lanes. Signage can sometimes be inadequate, requiring vigilance, especially for drivers unfamiliar with the region.

Alternative Routes for Emergencies

In case of roadblocks or emergencies, an alternative route through less-trafficked local roads may be necessary. It's advisable to plan these diversions in advance, considering back roads that bypass major bottlenecks or accident-prone areas. Using GPS applications for real-time updates is recommended.

Local Regulations on Hazardous Material Transport

Tamil Nadu has specific regulations on the transport of hazardous materials, requiring special permits and adherence to safety protocols. These regulations often stipulate restricted movement during nighttime or through specific urban areas. Compliance with these laws is critical to avoid legal issues and ensure safety.

Historical Incidents

While specific data is necessary for historical incidents, truck rollovers and fires involving heavy vehicles transporting hazardous materials have occurred in Tamil Nadu. These usually happen due to speeding, driver fatigue, or mechanical failure. Being aware of common causes and exercising caution can help prevent such incidents.

Environmental Considerations and Sensitive Areas

The route may pass through environmentally sensitive areas or regions with agricultural fields. Proper containment of hazardous materials is essential to avoid spills or contamination. Areas near water bodies or natural reserves demand extra caution.

Communication Coverage

Network coverage can vary, especially when transitioning from urban to rural regions. Notable dead zones typically occur in less populated stretches. Having alternative communication methods, such as satellite phones or radio communication, may be necessary in case of emergencies.

Emergency Response Times

Emergency response times can vary due to traffic and remoteness. Typically, urban centers have quicker response times (15-30 minutes), while rural or less accessible areas might experience delays (up to 45 minutes or longer). Establishing communication with local emergency services is advisable.

Overall Summary of Risk Assessment

The route from Sangagiri to Namakkal presents a typical risk profile for transporting hazardous materials. The primary concerns involve weather conditions during the monsoon season, traffic congestion, and road quality in less maintained sections. Adherence to regulations, proactive communication strategies, and emergency preparedness significantly mitigate potential risks. Given the infrastructure and geological considerations, the overall safety of the route is manageable with proper precautions and itinerary planning.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	Medium	11.38342, 77.89975	30 KM/Hr
6	Turn	Medium	11.38190, 77.90254	30 KM/Hr
7	Turn	High	11.38186, 77.90258	15 KM/Hr
8	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
9	Blind Spot	Blind Spot	11.23936, 78.14267	10 KM/Hr
10	Turn	High	11.23027, 78.13705	15 KM/Hr
11	Turn	High	11.23078, 78.13298	15 KM/Hr
12	Turn	Medium	11.22970, 78.13291	30 KM/Hr
13	Turn	Medium	11.22883, 78.13229	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
14	Turn	High	11.22868, 78.13200	15 KM/Hr
15	Turn	High	11.22332, 78.13129	15 KM/Hr
16	Turn	Medium	11.22332, 78.13131	30 KM/Hr
17	Turn	Medium	11.22385, 78.13244	30 KM/Hr
18	Turn	High	11.22384, 78.13250	15 KM/Hr
19	Turn	High	11.22469, 78.13303	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
11	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
12	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium
13	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium
14	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38342, 77.89975



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38190, 77.90254



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38186, 77.90258



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.37841, 77.90140



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.23936, 78.14267



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.23027, 78.13705



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.23078, 78.13298



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.22970, 78.13291



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.22883, 78.13229



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.22868, 78.13200



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.22332, 78.13129



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.22332, 78.13131



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.22385, 78.13244



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.22384, 78.13250



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.22469, 78.13303